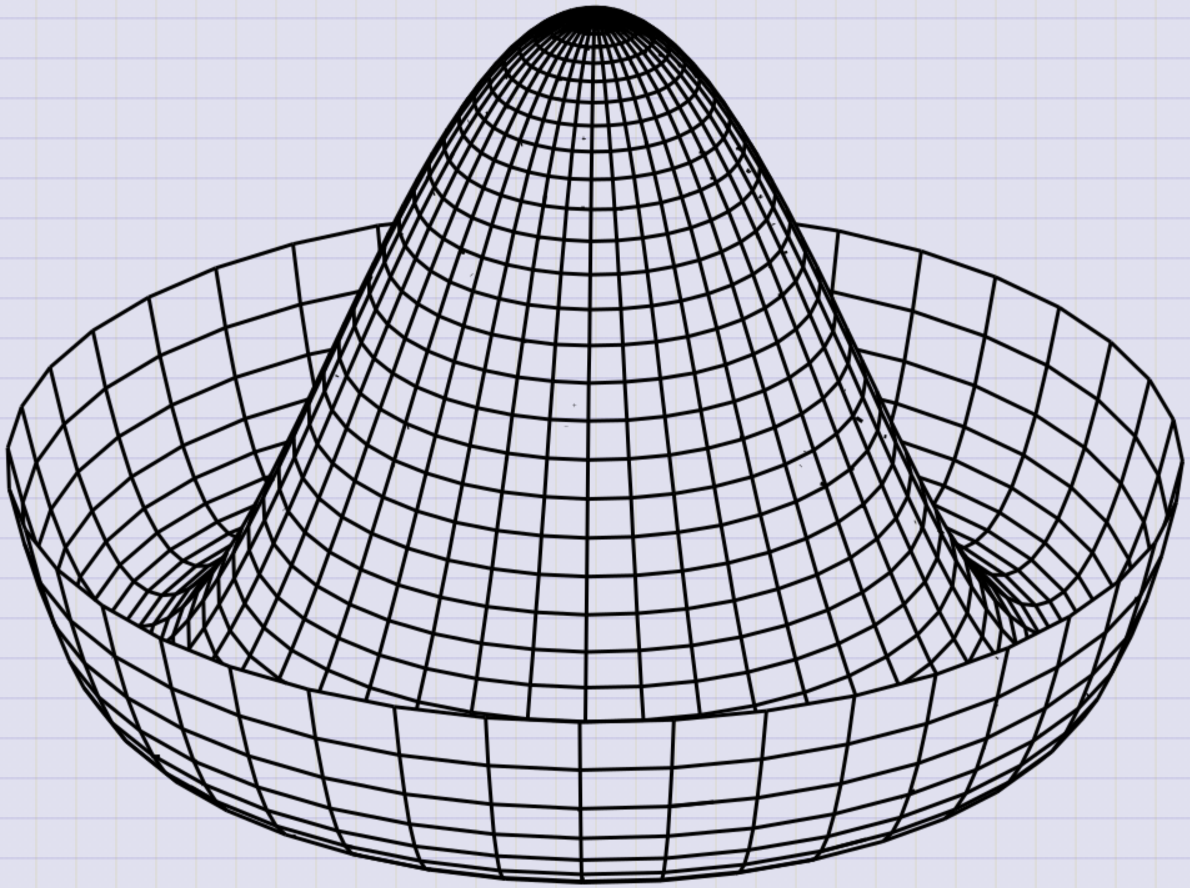
Day 15 - Potential Energy and Stability

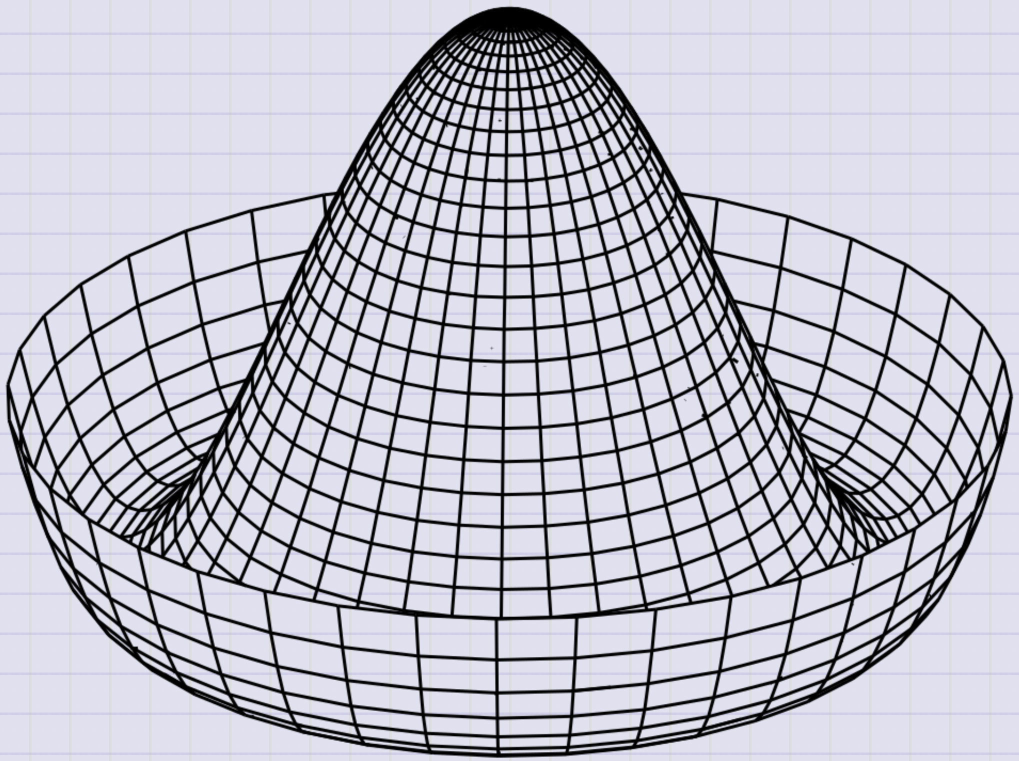
Mexican Hat/Sombrero Potential $\longrightarrow$



Mexican Hat Potential

$V(\phi) = -5|\phi|^2 + |\phi|^4$

- Spontaneous Symmetry Breaking (Jeffery Goldstone, 1961)
- Unstable vacuum state at $\phi = 0$
  - Peak of the hat
- Infinite number of stable minima
  - $\phi = \sqrt{5/2}e^{i\phi}$



Announcements

- Midterm 1 is available today (Due Oct 10th)
- You may work in larger groups, but solutions are submitted like homework
- No office hours on Friday (DC traveling)

Seminars this week

MONDAY, September 29, 2025

Condensed Matter Seminar 4:10 pm,1400 BPS, In Person and Zoom, Host ~Mark Dykman

Speaker: Peter Littlewood, University of Chicago

Title: Non-Reciprocal Phase Transitions

Zoom Link: https://msu.zoom.us/j/93613644939

Meeting ID: 936 1364 4939

Password: CMP

Seminars this week

TUESDAY, September 30, 2025

Theory Seminar, 11:00am., FRIB 1200 lab in person and online via Zoom

Speaker: Debora Mroczek, University of Illinois

Title: Revealing new phases of matter in neutron stars

Please click the link below to join the webinar:

Zoom Link: 964 7281 4717

Meeting ID: 48824

Seminars this week

TUESDAY, September 30, 2025

High Energy Physics Seminar, 1:30 pm, 1400 BPS, Host~ Joshua Isaacson

Speaker: Tony Menzo, Joint affiliation - FNAL/Univ. of Alabama

Title: The journey towards differentiable hadronization models

Zoom:

Passcode:

(Joining the Zoom meeting requires a password. Please contact one of the organizers, if you haven't received it.)

Organized by: Joey Huston, Sophie Berkman and Brenda Wenzlick

Seminars this week

WEDNESDAY, October 1, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Madison Brady, MSU

Title:

Zoom Link: https://msu.zoom.us/j/93334479606?pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1

Meeting ID: 933 3447 9606

Passcode: 825824

Seminars this week

THURSDAY, October 2, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~ Witold Nazarewics

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: David Dean, Thomas Jefferson National Accelerator Facility

Title: From the quarks to the cosmos

Background:

For more information and to schedule time with the speaker, see the colloquium calendar at https://pa.msu.edu/news-events-seminars/colloquium-schedule.aspx

Zoom Link: https://msu.zoom.us/j/94951062663

Password: 2002

Seminars this week

FRIDAY, October 3, 2025

Special Seminar, 10:00am, In Person Only, FRIB 1300 Auditorium

Speaker: Sudip Bhattacharyya - Tata Institute of Fundamental Research & Massachusetts Institute of Technology

Title: Thermonuclear X-ray bursts: probing neutron stars and a double-photospheric-radius-expansion

Sign up to meet with Dr. Bhattacharyya (enter your name and meeting location): https://docs.google.com/spreadsheets/d/1tlw765Lf8mNOeCUJp8cWfSOvQfiF4vJDOuhKKVPXH1M/edit?usp=sharing

Seminars this week

FRIDAY, October 3, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, Zoom only

Speaker: Balint Pato, Duke University

Title: Compass codes in quantum error correction

Full Scheule is at: https://sites.google.com/msu.edu/quic-seminar/

For more information, reach out to Ryan LaRose

This Week's Goals

- Understand the concept of potential energy
- Determine the equilibrium points of a system using potential energy
- Analyze the stability of equilibrium points
- Define and begin to apply conservation of linear and angular momentum

Reminder: The Gradient Operator ∇

∇ is a vector operator. In Cartesian coordinates:

$$\nabla = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

Acting on a scalar function $f(x, y, z)$ produces a vector:

$$\nabla f(x, y, z) = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z} = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Reminder: The Gradient Operator ∇

∇ can act on vector field (function), $\mathbf{F}(x, y, z)$ with both dot and cross products.

Divergence (Scalar Product)

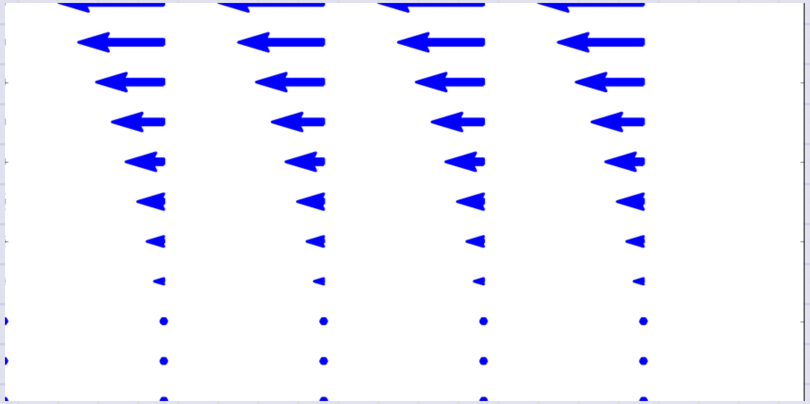
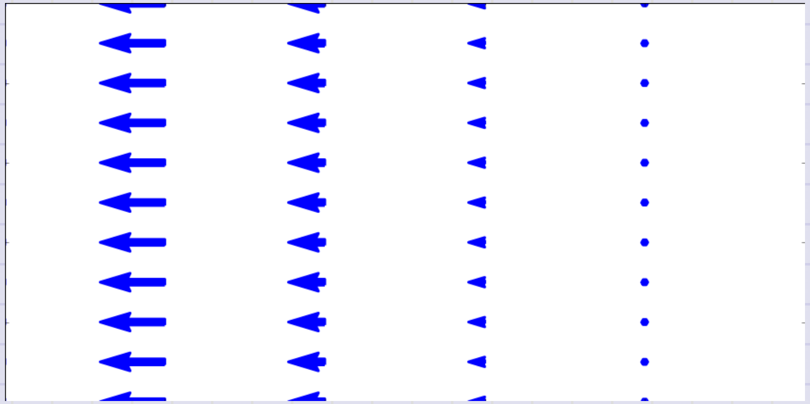
$$\nabla \cdot \mathbf{F}(x, y, z) = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle F_x, F_y, F_z \rangle$$

$$\nabla \cdot \mathbf{F}(x, y, z) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

How does the vector field change in the direction of the vector?

Clicker Question 15-1a

Which of the following fields have no divergence?

A.  B. 

1. A
2. B
3. Both A and B
4. Neither A nor B

Reminder: The Gradient Operator ∇

∇ can act on vector field (function), $\mathbf{F}(x, y, z)$ with both dot and cross products.

Curl (Vector Product)

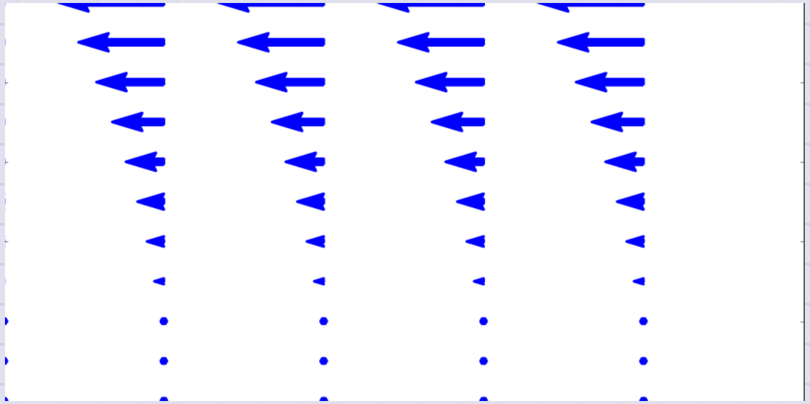
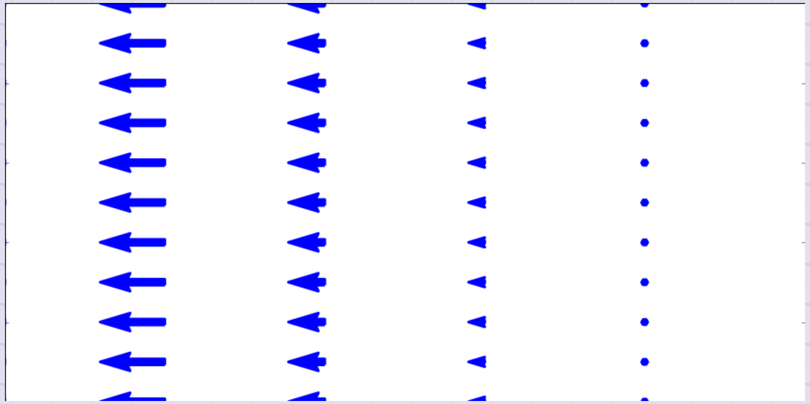
$$\nabla \times \mathbf{F}(x, y, z) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$\nabla \times \mathbf{F}(x, y, z) = \left\langle \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right\rangle$$

How does the vector field change in the direction perpendicular to the vector?

Clicker Question 15-1b

Which of the following fields have no curl?

A.  B. 

1. A
2. B
3. Both A and B
4. Neither A nor B

Clicker Question 15-1c

Consider a vector field with zero curl: $\nabla \times \vec{F} = 0$. Which of the following statements is true?

1. The field is conservative
2. $\int \nabla \times \vec{F} \cdot d\vec{A} = 0$
3. $\oint \vec{F} \cdot d\vec{r} \neq 0$
4. $\vec{F}$ is the gradient of some scalar function, e.g., $\vec{F} = -\nabla U$
5. Some combination of the above

Reminders: Conservative Forces

- Conservative forces are those with zero curl

$$\nabla \times \vec{F} = 0$$

- The work done by a conservative force is path-independent; on a closed path, the work done is zero

$$\oint \vec{F} \cdot d\vec{r} = 0$$

- The force can be written as the gradient of a scalar potential energy function

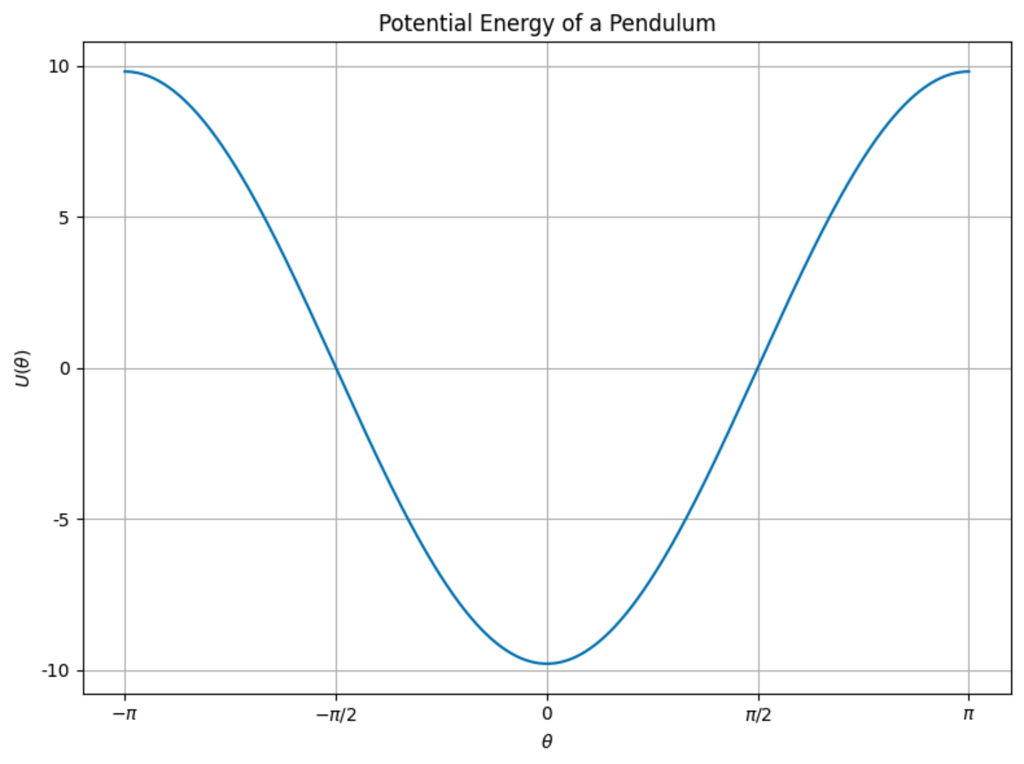
$$\vec{F} = -\nabla U$$

Clicker Question 15-2

Here's the graph of the potential energy function $U(x)$ for a pendulum.

What can you say about the equilibrium points? There is/are:

1. One stable point
2. Two stable points
3. One stable and one unstable point
4. Two unstable and one stable point



Clicker Question 15-3

Here's a potential energy function $U(x)$ for a pendulum:

$$U(\phi) = -mgL\cos(\phi) + U_0$$

1. Find the equilibrium points (ϕ^*) of the pendulum by setting:

$$\frac{dU(\phi^*)}{d\phi} = 0.$$

2. Characterize the stability of the equilibrium points (ϕ^*) by examining the second derivative:

$$\frac{d^2U(\phi^*)}{d\phi^2} > 0? \qquad \frac{d^2U(\phi^*)}{d\phi^2} < 0?$$

Click when done.

Clicker Question 15-4

A double-well potential energy function $U(x)$ is given by

$$U(x) = -\frac{1}{2}kx^2 + \frac{1}{4}kx^4.$$

We assume we have scaled the potential energy so that all the units are consistent.

How many equilibrium points does this system have?

1. 1
2. 2
3. 3
4. 4

Clicker Question 15-5

A double-well potential energy function $U(x)$ is given by

$$U(x) = -\frac{1}{2}kx^2 + \frac{1}{4}kx^4.$$

1. Find the equilibrium points (x^*) of the pendulum by setting:

$$\frac{dU(x^*)}{dx} = 0.$$

2. Characterize the stability of the equilibrium points (x^*) by examining the second derivative:

$$\frac{d^2U(x^*)}{dx^2} > 0? \qquad \frac{d^2U(x^*)}{dx^2} < 0?$$

Click when done.

Clicker Question 15-6

Here's a graph of the potential energy function $U(x)$ for a double-well potential.

Describe the motion of a particle with the total energy, $E =$

1. $0.4\,\mathrm{J}$, $<$ barrier height
2. $1.2\,\mathrm{J}$, $>$ barrier height
3. $1.0\,\mathrm{J}$, $=$ barrier height

Click when done.